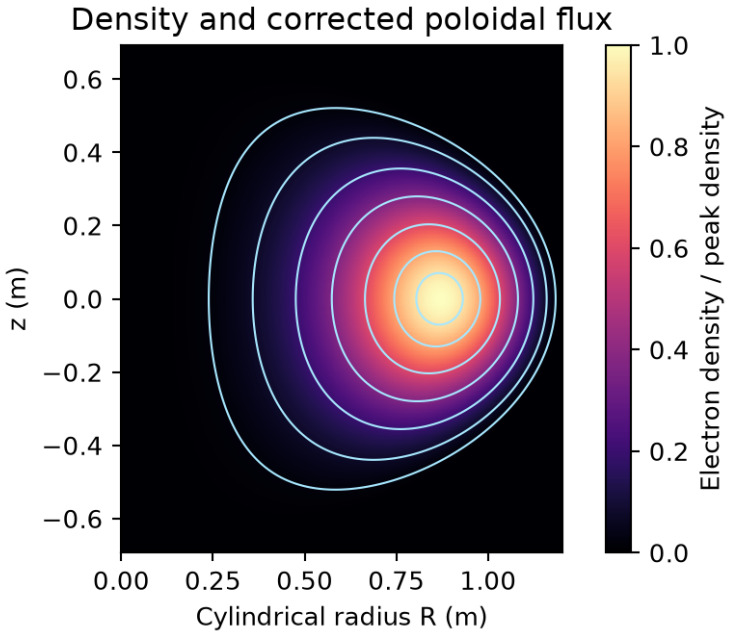
The calculated organization

An oblate domain. Opposing mean circulation. A nonzero electric current.

This 45-second film makes an existing stationary waterbag construction visible. Its key result is exact in that construction: the summed mechanical angular momentum cancels pointwise, while the electron, deuterium and tritium currents reinforce one another.



Actual geometry. $R^2/a^2 + z^2/c^2 \le 1$, with a = 1.202057 m, c = 0.692070 m, $q = (a/c)^2 = 3.016827$ and volume 4.188790 m^3. The selected member has peak B = 5 T and peak electron temperature parameter 16.71 keV.

Regeneration, not invented field lines. The archived degree-nine conditioned constructor regenerated χ and density with relative L^2 differences of 0.0 against the stored 2,304-point profile in this run. The historical strong-field residual is 0.0744%. The rendering uses a new 641 × 641 R–z field grid and contours of the corrected flux ψ.

The density peaks away from the axis and vanishes at the boundary. Its bright annular region is a property of this computed member; no toroidal vessel or external coil geometry is inserted. The lavender curves show the admitted interior magnetic field only.

Scope. Numerical visualization of a stationary, collisionless, quasineutral construction. It is not reactor footage, a stability demonstration or an integrated burning-plasma return.

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Why current survives

Charge-weighted circulation and mass-weighted angular momentum are different accounts.

$$f_s = C_s \mathbf{1}_{H_s - \Omega_s P_{\phi s} < 0}, \quad P_{\phi s} = m_s R v_\phi + q_s \psi$$

$$n_D = n_T = \tfrac{1}{2} n_e, \quad \Omega_e = -\Omega, \quad \Omega_D = \frac{m_e}{m_D}\Omega, \quad \Omega_T = \frac{m_e}{m_T}\Omega$$

$$\ell_z = R^2 \sum_s m_s n_s \Omega_s = m_e n_e \Omega R^2 (-1 + \tfrac{1}{2} + \tfrac{1}{2}) = 0$$

$$J_\phi = e n_e \Omega R \left(1 + \frac{m_e}{2m_D} + \frac{m_e}{2m_T}\right) > 0$$

Species	Rotation / rad s^{-1}	Angular momentum / kg m^2 s^{-1}
Electron	−65,743.1188568	$-2.07095391846 \times 10^{-4}$
Deuterium	+17.9113137401	$+1.03547695923 \times 10^{-4}$
Tritium	+11.9599919331	$+1.03547695923 \times 10^{-4}$
Sum	Not an angular velocity	0

Mass ratios in the archived calculation: m_D/m_e = 3670.48 and m_T/m_e = 5496.92. The movie retains those rotation-rate ratios exactly. The tracers represent mean circulation, not the full thermal velocity of individual particles.

$$-\Delta^* \hat{\chi} = \kappa \hat{R}^2 \hat{\chi}^{3/2}, \quad \hat{\chi} = \hat{R}^2 u, \quad \hat{\psi} = \hat{\chi} - \tfrac{1}{2} h \hat{R}^2$$

$$B_R = -B_* \hat{R} u_{,\hat{z}}, \quad B_z = B_* (2u + \hat{R} u_{,\hat{R}} - h), \quad B_\phi = 0$$

$$n_e \propto \chi^{3/2}, \qquad p \propto \chi^{5/2}$$

The uniform centrifugal bias is retained: −0.373791 μT in B_z. Zero net mechanical angular momentum does not mean zero individual rotation, zero centrifugal stress, or zero electromagnetic angular momentum.

How to read and reproduce it

The numerical receipt and the cinematic treatment remain separately inspectable.

One clock, two viewing scales. 0–13 s: 40 μs of physical mean-flow time per film second. After 13 s: 75 ms per film second. All species use the same clock; positions are continuous across the change.

Exposure, not altered ion speed. At the ion scale, fast electron circulation becomes a full-turn exposure average. Twenty-five representative cyan rings remain visible; the density volume retains the electron spatial profile. No separate speed multiplier is applied to fuel ions.

A visible model, not an optical prediction. Cyan identifies electrons, amber deuterium, rose tritium and lavender calculated magnetic curves. Brightness, glow and camera lighting are illustrative. Density determines the spatial volume. Original ambient audio is sound design, not measured plasma sound.

Exact source trail. Copied constructors: edge_regular_ritz.py → kinetic_constructor.py → conditioned_fields.py. Parameter receipt: Burn_Renewal/results/dimensional_results.json, q = 3.016827, degree 9, L = 1 m. Original research: R. Wade H. Marr and Joseph Adam Thiel.

Machine-readable companion. TECHNICAL_COMPANION.md retains literal LaTeX. data/visualization_receipt.json records checks and source hashes. field_grid.npz, field_lines.json and tracers.npy contain the actual visual inputs.

Reproduction. Run python production/regenerate_data.py; then Blender 4.3 with -b --python production/scene.py -- --render; then python production/finish_film.py. README.md contains the complete commands and environment. The original research archive remains unchanged.

The exact cancellation belongs to the selected waterbag construction. This visualization does not establish collisional sustainment or reactor performance. Its purpose is to expose the difference between current and net mechanical spin through the actual constructed geometry and circulation.